

1A

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ |
```

1B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ |
```

2A and 2B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ make test_objects
```

3A

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "hello" > file1.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "world" > file2.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes add file1.txt file2.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: .gitignore
  untracked: commit.c
  untracked: commit.h
  untracked: file.txt
  untracked: fix.py
  untracked: fix2.py
  untracked: fix_index_add.py
  untracked: index.c
  untracked: index.h
  untracked: Makefile
  untracked: object.c
  untracked: pes.c
  untracked: pes.h
  untracked: README.md
  untracked: solve.py
  untracked: solve2.py
  untracked: test_objects.c
  untracked: test_sequence.sh
  untracked: test_tree.c
  untracked: tree.c
  untracked: tree.h
```

3B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776343867 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776343874 6 file2.txt
```

4A

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "hello" > hello.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes add hello.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes commit -m "Initial commit"
Committed: d2f66317b637... Initial commit
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "World" >> hello.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes add hello.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes commit -m "Add world"
Committed: d1b0bc424577... Add world
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "Goodbye" > bye.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes add bye.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes commit -m "Add farewell"
Committed: 29c1e3d5e493... Add farewell
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes log
commit 29c1e3d5e493be7ed6f56c22fc8d80b57a49281b938bf84bbb9ba0c4c634fe87
Author: PES User <pes@localhost>
Date: 1776346898

    Add farewell

commit d1b0bc4245776ea6ab4eba82938ade00774db1b1e78b9c882b674641ff200ff5
Author: PES User <pes@localhost>
Date: 1776346877

    Add worldell

commit d2f66317b6371d787b4831562ee96bcc3dbc5cddb426e9620f2b826eab37592b
Author: PES User <pes@localhost>
Date: 1776346849

    Initial commit

commit 4c305daef0b899e311d623fa8f3eda5f8131fd527c1e6d2dcaabd9a73a05ebbd
Author: PES User <pes@localhost>
Date: 1776344899

message9a
```

4B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/17/b4c616f8c09f4ff43f89cefe3b863fad4fe8cfec19f921b8829ef8e07750bc
.pes/objects/1a/db013adb523ffd3f4fe85f6a7b6c6f91ea20571455d23e5904c661237f7871
.pes/objects/29/c1e3d5e493be7ed6f56c22fc8d80b57a49281b938bf84bbb9ba0c4c634fe87
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/4c/305daef0b899e311d623fa8f3eda5f8131fd527c1e6d2dcaabd9a73a05ebbd
.pes/objects/5d/6a4759aa7262e1d3e9b2b778f0feb43be999a4867294379064bd0a89b1d3cf
.pes/objects/a9/9e50f4e6affe4383692c1ed3b6418cc131583f2a46245a20419009cb2d160d
.pes/objects/b4/e76aee38854839557e5a3132e346adb8f8288ec2ba878f2cd915d5c2326af00
.pes/objects/d1/b0bc4245776ea6ab4eba82938ade00774db1b1e78b9c882b674641ff200ff5
.pes/objects/d2/f66317b6371d787b4831562ee96bcc3dbc5cddb426e9620f2b826eab37592b
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ cat .pes/refs/heads/main
```

4C

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ cat .pes/refs/heads/main
29c1e3d5e493be7ed6f56c22fc8d80b57a49281b938bf84bbb9ba0c4c634fe87
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ |
```

Phase 5 and 6

5.1

Update HEAD to point to .pes/refs/heads/<branch>

Read commit hash from branch file

Load commit -> get tree -> restore all files in working directory

5.2

For each tracked file:

Get hash from index

Compute hash of working file

If hashes differ -> file is modified

If modified file also differs in target branch -> refuse checkout

5.3

HEAD points directly to a commit

New commits are created but **not linked to any branch**

They may become unreachable

Recovery:

Create a branch using commit hash

6.1

Start from all branch heads

Traverse commits -> trees -> blobs

Mark reachable objects (use **HashSet**)

Delete unvisited objects

Estimate:

Need to visit hundreds of thousands of objects

6.2

During commit, new objects may not yet be referenced

GC may delete them -> commit becomes invalid

Git avoids this by:

using locks

delaying deletion (grace period)

ensuring atomic updates